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EMPIRICAL PAPER

Shedding light on the effects of supportive techniques on nonverbal synchrony and their moderators in psychotherapy for depression

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Abstract

Objective: Accumulating research demonstrates the importance of utilizing supportive techniques in psychotherapy; however, little is known about therapeutic processes that are set in motion following the use of supportive techniques. The present study examined the effects of supportive techniques on nonverbal synchrony, both at the sample level and at the individual differences level.

Method: The sample consisted of 86 patients from a randomized controlled trial for treatment of depression. Supportive techniques were rated by patients and therapists after every session, and nonverbal synchrony was quantified by motion energy analysis (MEA) for each session. The ability of supportive techniques to predict subsequent nonverbal synchrony was examined using polynomial regression and response surface analysis.

Results: The findings suggest that, at the sample level, greater use of supportive techniques was a significant predictor of subsequent higher levels of nonverbal synchrony. At the individual differences level, this effect was significant for patients with low levels of depression severity and personality disorders, yet not significant for patients with high levels.

Conclusion: The present study demonstrates that greater use of supportive techniques in treatment may facilitate a process that manifests as higher levels of synchrony, especially for patients with lower levels of personality disorders and depression.

Keywords: supportive techniques; nonverbal synchrony; therapeutic relationship; supportive-expressive treatment; psychotherapy process

Clinical or methodological significance of this article: The findings of the present study suggest that the use of supportive techniques within a therapy setting may influence and change the nonverbal synchrony between patient and therapist at the subsequent session. This finding was true for patients with low levels of depression severity and personality disorders, not for patients with high levels. Therefore, this study stresses the importance of understanding how supportive techniques may facilitate a process in psychotherapy that manifests as higher levels of synchrony, taking into account individual differences between patients.

In recent decades, there has been increasing acknowledgment of the importance of supportive techniques in psychotherapy (Appelbaum, 2006; De Jonghe et al., 2004). Supportive techniques are a broad term that refers to therapists' actions that aim to facilitate a therapeutic environment for the patient, to foster and strengthen the relationship between the patient and therapist and help bring them to a shared experience (Barber et al., 2001; De Jonghe et al., 1994; Wachtel, 2011). Supportive techniques are considered to facilitate the patient in feeling more secure and free to express thoughts

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and feelings that may be difficult to share, which contributes to the closeness between the patient and therapist and to their collaborative work (Luborsky, 1984; Misch, 2000). Supportive techniques encourage the therapist to focus in alignment with the patient on the here-and-now (Book, 1998) and work together as a team to facilitate better self-understanding (Barber & Crits-Christoph, 1996). The literature refers to a wide range of supportive techniques, including demonstrating empathy, acceptance and affection; noting the patient's gains; believing in the patient's strengths and acknowledgment of the growing ability of the patient to find solutions without the therapist's help (Barber & Crits-Christoph, 1996; Luborsky, 1984).

Previous studies suggest that treatments which are based on supportive techniques succeed in significantly reducing symptoms of depression (Cuijpers et al., 2012), as efficaciously as other types of techniques (Driessen et al., 2010). Furthermore, it has been suggested that supportive techniques can be considered a common therapeutic intervention across different therapies (Barber et al., 2001; Wachtel, 2011). Despite the growing number of studies on the merit of supportive techniques (de Maat et al., 2008), little is known empirically about the therapeutic processes that are initiated following the use of supportive techniques in therapy. It has been suggested that the ability of the patient and therapist to be in a shared experience may be one factor that is set in motion following the use of supportive techniques.

One manifestation of the ability of two interacting partners to be in a shared experience is the level of synchrony between them (Koole et al., 2020). A prolific body of research has shown that synchrony between interacting partners is a fundamental mechanism within interpersonal interactions (Chartrand & Lakin, 2013; Vicaria & Dickens, 2016). Synchrony between interacting partners was found to be linked to positive and beneficial social connections, including feelings of closeness, rapport and collaboration (Wiltermuth & Heath, 2009), a common understanding, empathy, emotional sharing and regulation (Feldman, 2012; Koole & Tschacher, 2016). The positive and beneficial social connections associated with synchrony are manifested within various types of relationships: relations between people in groups (Chartrand & Lakin, 2013), parent–child dyads (Feldman, 2012) and romantic partners (Julien et al., 2000). Following the great interest in synchrony in basic science research, recently there has been a growing focus on synchrony within psychotherapy research (Koole et al., 2020; Wiltshire et al., 2020). One of the promising approaches to examining synchrony in psychotherapy is the study of nonverbal synchrony between patients and therapists (Ramseyer & Tschacher, 2011). It is defined as the coordination of movement between interacting partners, irrespective of their postures or gestures (Condon & Ogston, 1966; Ramseyer & Tschacher, 2011). Empirical findings show that higher levels of nonverbal synchrony are associated with an improved therapeutic relationship between the patient and therapist, favorable treatment outcomes, and greater symptom reduction (Altmann et al., 2020; Ramseyer & Tschacher, 2011).

The literature shows the benefits of supportive techniques on the one hand, and the benefits of being in high levels of nonverbal synchrony on the other hand, and that theoretically they have a lot in common. However, little empirical research has connected these two theoretically related constructs. To the best of our knowledge, only one study has focused on therapists' interventions and their association with nonverbal synchrony, reporting promising results (Prinz et al., 2021). In this study, Prinz et al. (2021) found that higher levels of same-session nonverbal synchrony were associated with greater use of "mastery interventions," such as supporting the patient to learn coping and problem-solving strategies within a supportive, active, and engaging interaction. By contrast, greater use of "resource activation," such as focusing on patients' strengths, was found to be negatively associated with nonverbal synchrony within patient-therapist dyads. This study contributes to our understanding of the ability of nonverbal synchrony to predict therapists' interventions at the same session, but two important issues remain unanswered: (a) the ability of therapists' techniques to predict nonverbal synchrony at a subsequent session, and (b) taking into account both patients' and their therapists' ratings of the use of techniques. The level of congruence between the patient and therapist ratings of various aspects of therapy has been shown to be associated with better process and outcome of treatment (Chui et al., 2020; Marmarosh & Kivlighan Jr, 2012; Rubel et al., 2018). Thus, the first aim of the present study is to examine whether, at the sample level, greater use of supportive techniques, as rated congruently by both patient and therapist, predicts subsequent higher levels of nonverbal synchrony.

While at the sample level, we expect greater use of supportive techniques to predict subsequent higher levels of nonverbal synchrony, variability may emerge between individuals. Patients' personal factors, including intra- and interpersonal characteristics, have been found to explain individual differences in therapeutic processes (Bohart & Wade, 2013). Among patients with major depressive disorder (MDD), factors of great interest are those

that might impair the ability of the patient and therapist to be in a shared experience, and engage in a close and collaborative interaction. Specifically, the literature refers to depression severity and personality disorder (PD) levels (Clarkin & Levy, 2004) as patients' personal factors that may impact therapeutic processes in relation to difficulties in forming and maintaining interpersonal relationships.

Although little is known of the effects of depression severity and PD levels on the ability of the patient and therapist to be in a shared experience and synchronize, these two factors have been shown to be consistently associated with interpersonal difficulties (Benjamin, 1993; Livesley, 2001; Pearson et al., 2010). In therapy, numerous studies have demonstrated that higher levels of patients' symptom severity at the beginning of a session predicted a weaker therapeutic relationship between patient and therapist in the same session (for review, see Flückiger et al., 2020). Similarly, previous studies found that patients with high levels of PD are more likely to have an unstable relationship with their therapists, characterized by moment-to-moment changes and shifts, unlike patients with low levels of PD (Levy et al., 2010; Muran et al., 2009; Strauss et al., 2006). In view of these findings, patients' depression severity and high levels of PD may impair the ability of the patient and therapist to be in a shared experience and synchronize. Thus, the second aim of the present study is to examine whether the effect of supportive techniques on subsequent levels of nonverbal synchrony is moderated by patient factors, specifically depression severity and PD levels.

In the present study, we sought to examine the processes that are set in motion following the use of supportive techniques in therapy. Specifically, we examined whether supportive techniques can predict subsequent higher levels of nonverbal synchrony, in a data set in which we previously have shown that nonverbal synchrony was associated with a better therapeutic relationship between the patient and therapist (Cohen et al., 2021). To yield a more dependable assessment of the techniques used in the sessions, we used both the patient and therapist ratings, rather than using only therapist or patient ratings. We assessed the level of congruence between them using polynomial regression and response surface analysis (Edwards, 2002; Edwards & Parry, 1993). According to the response surface analysis, congruence and incongruence are not the same thing in psychotherapy: one focuses on cases in which the two partners are *congruent* and their ratings are both high or low, whereas the other focuses on cases in which they are *incongruent* and examines the sources of such incongruence (Marmarosh & Kivlighan Jr, 2012). In the present study, we examined the effect of congruence and incongruence between patients and therapists regarding the use of supportive techniques, on subsequent nonverbal synchrony at (a) the sample level, and (b) the individual differences level, specifically, patients' depression severity and PD levels. We hypothesized that, at the sample level, patient and therapist congruence on higher levels of supportive techniques would be associated with subsequent higher levels of nonverbal synchrony. At the individual differences level, despite the scarcity of the literature directly examining this issue, we expect the effect of supportive techniques on subsequent nonverbal synchrony to be more associated with patients with low levels of depression severity or low levels of PD, and less associated with patients with high levels. The analyses of the nature of the effect (congruence and incongruence) at the individual differences level were exploratory, because of the lack of previous research on this topic. To achieve these aims, we repeatedly assessed the supportive techniques and nonverbal synchrony of the dyad over the course of 16 sessions of an ongoing randomized controlled trial (RCT) of 86 patients with MDD.

Method

Participants

Eighty-six patients with MDD were recruited through advertisements offering free treatment for depression. This study was part of a larger ongoing RCT comparing supportive and supportive-expressive therapy for MDD (for more details see Zilcha-Mano et al., 2018). Inclusion criteria were: (a) meeting MDD diagnostic criteria based on structured clinical interviews for DSM-V, with scores above 14 on the 17-item Hamilton Rating Scale for Depression (HRSD; Hamilton, 1967) at two evaluations, one week apart, and current MDD based on the Mini-International Neuropsychiatric Interview (MINI; Sheehan et al., 1998); (b) if the patients were on medication, their dosage must have been stable for at least three months before entering the study, and they had to be willing to maintain a stable dosage for the duration of the treatment; (c) age between 18 and 60 years; (d) Hebrew language fluency; (e) provision of written informed consent. Exclusion criteria were: (a) current risk of suicide or self-harm (HRSD suicide item >2); (b) current substance abuse disorder; (c) current or past schizophrenia or psychosis, bipolar disorder, or severe eating disorder requiring medical monitoring; (d) history of organic mental disease; (e) currently in psychotherapy.

The mean age of the participants was 32.03 years ($SD = 8.54$), and the majority were female (59.3%). Of the participants, 73.3% were single, 19.7% married or in a relationship, 4.7% divorced or separated, 1.2% widowed, and 1.2% indicated "other." The mean level of education was 14.83 years ($SD = 2.904$). In addition, 47.7% were employed, 39.5% were students, 9.3% were unemployed, 2.3% were homemakers, and 1.2% indicated "other." At intake, all patients met criteria for a primary diagnosis of MDD, and 70.9% of the patients were diagnosed with one or more personality disorders. The most frequent personality disorders were obsessive-compulsive (46.5%), avoidant (30.2%), dependent (16.3%), borderline (15.1%), narcissistic (12.8%), and histrionic (5.8%).

Therapists

Nine therapists, with at least five years of expertise in psychodynamic treatment, participated in the study. All had formal training and experience in psychodynamic treatment. The therapists attended a 20-hour training workshop in supportive and expressive techniques before seeing patients. All therapists completed treatment of two pilot patients, one of each treatment type, and had to demonstrate sufficient adherence before moving to the trial phase. Throughout the study, the therapists received weekly personal and group supervision, provided by two experienced licensed clinical psychologists, who themselves received supervision on supervision from a clinician with more than 20 years of experience in psychodynamic treatment for depression, and more than 15 years of experience in supportive-expressive therapy (SET) in RCTs. Therapists provided both treatment conditions, to act as their own controls and avoid nesting of therapists within treatment conditions, which may result in unwanted confounding. Mean clinical experience of the therapists was 11.88 years ($SD = 5.72$), mean age was 39.88 ($SD = 6.15$), and 66.7% were women. The mean number of patients treated by each therapist in the current study was 9.55 ($SD = 8.202$; range = 1–19).

Treatment

Patients received 16 weekly 50-minute sessions of SET, a time-limited psychodynamic therapy adapted for depression, either in an expressive-focused condition or in a supportive-focused one. Assignment to treatment condition was conducted by an outside institution, not involved in the study. Following the general requirement in psychotherapy research not to break the blindness to conditions before the end of the RCT, in this study, as in other studies in the literature, the two conditions were analyzed together. We used comprehensive treatment protocols for SET: the Luborsky manualized treatment (Luborsky, 1984; Luborsky et al., 1995). The supportive condition included all supportive techniques detailed in the manual, but forbid the use of any expressive techniques (as detailed in Leibovich et al., 2018).

Measures

Supportive techniques. The use of supportive techniques was measured with the common factor subscale in the Multi-theoretical List of Therapeutic Interventions (MULTI-30; McCarthy & Barber, 2009; Solomonov et al., 2019). This is a 30-item self-reported measure assessing intervention use from eight therapy orientations (cognitive, behavioral, process-experiential, person-centered, psychodynamic, interpersonal, dialectical-behavioral, and common factors) from the perspective of both patient and therapist. In the present study, we focused on the common factor subscale, which assesses positive expectancies and relationship factors associated with supportive treatment, such as warmth, acceptance, and attunement. Items are rated on a 5-point Likert scale, rating each item based on how representative it is of the session, on a scale ranging from 1 (Not at all) to 5 (Very). We used 16 session-by-session measures of the patient's and therapist's versions of the MULTI-30 common factor subscale. In the current study, the internal reliability of the MULTI-30 common factor subscale was 0.72 for the patient version and 0.58 for the therapist version.

Personality disorders. The patient's personality disorders were measured using the Structured Interview for the Diagnosis of Personality Disorders (SIDP-IV; Pfohl et al., 1997). This instrument is a comprehensive semi-structured clinical interview for the assessment of DSM-IV Axis II diagnoses. The SIDP-IV includes non-pejorative questions organized into topical sections to produce a natural flow in the interview. The number of criteria for each PD in the DSM-IV varies from 7 to 9. Criteria were rated as follows: 0=Absent, 1=Sub-threshold, 2=Present, 3=Strongly present. A score of 2 or more on at least 3–5 criteria (depending on the PD in question) is required for a diagnosis of PD. The instructions for the SIDP-IV specify a scoring rule, according to which behavior typical of the past 5 years represents the basis for the ratings. SIDP-IV

interviewers were masters or doctoral level clinical psychologists, who received extensive training and supervision in the administration of the SIDP-IV. Inter-judge reliability for the 79 items of the SIDP-IV, assessed by intraclass correlation (ICC; Shrout & Fleiss, 1979), was .93, which is considered excellent (Fleiss et al., 2003). PDs were examined both as a categorical and a continuous variable. A continuous assessment of PD was determined by tallying the total number of personality traits endorsed across all SIDP-IV items.

Severity of depressive symptoms. We assessed symptom severity with the Hamilton Rating Scale for Depression (HRSD; Hamilton, 1967), a 17-item clinically administered semi-structured interview, with higher scores indicating greater severity of depression. The HRSD interviewers received extensive training and supervision in the administration of the instrument. Inter-judge reliability for the current study, assessed by intraclass correlation (ICC; Shrout & Fleiss, 1979), was .96, which is considered excellent (Fleiss et al., 2003).

Nonverbal synchrony. Nonverbal synchrony of the dyad was measured using motion energy analysis (MEA; Ramseyer, 2020). We analyzed a total of N=1354 videotaped 50-min therapy sessions: 16 sessions for 86 patients. Twenty-two sessions were not available for analysis because of technical reasons or premature termination. Motion energy was defined as the difference in pixel values between consecutive video frames. The detection of frame-by-frame changes enables quantification of changes in predefined regions of interest (ROI). The ROIs of the head and body regions were summed up into one region per partner. Calculations of nonverbal synchrony were conducted using the rMEA R-package (Kleinbub & Ramseyer, 2021). The quantification of nonverbal synchrony was based on a time-lagged cross-correlation algorithm, which calculates the association of the two partners' time-series in a range of ±5 s. The mean of all absolute correlation values was considered to be the quantity of coordinated movements between the two partners; that is, a single-value representation of the global nonverbal synchrony between patient and therapist (Ramseyer, 2020; Ramseyer & Tschacher, 2011). We examined nonverbal synchrony as described above in contrast to pseudosynchrony, i.e., nonverbal synchrony caused by random coincidence, to obtain an estimate of the strength of the phenomenon of nonverbal synchrony (Ramseyer, 2020; Ramseyer & Tschacher, 2010). Nonverbal synchrony was adjusted by assessing a z-transformed variable by providing an effect-size estimate of nonverbal synchrony compared with pseudosynchrony (Ramseyer & Tschacher, 2011), which was used in all subsequent analyses (see the Supplements for more details).

Procedure

For baseline assessments, participants were invited to participate in three assessment meetings. At the first two meetings, we checked inclusion and exclusion criteria, explained the research goals, and provided information about participation in the research project. Patients willing to participate signed informed consent forms, confirming their understanding that all treatment sessions were to be videotaped, and that they had the right to withdraw from the research at any time. Patients were also told that their anonymity would be preserved. Subsequently, we conducted interviews using the HRSD-17 (Hamilton, 1967) and the MINI (Sheehan et al., 1998) to confirm the presence of a depressive disorder and to determine its severity. Patients with a MINI diagnosis of depressive disorder and HRSD score of 14 or above were included in the study. At the third meeting, we evaluated PDs using the SIDP- IV interview (SIDP-IV; Pfohl et al., 1997). After the three meetings, the 16-session therapy began. To assess the therapist's use of techniques, patients and therapists completed the MULTI (McCarthy & Barber, 2009; Solomonov et al., 2019) after every session. All research materials were collected after securing the approval of the Internal Review Board of the institution.

Data Analysis

The data were hierarchically nested on three levels: assessments nested within patients (16 observations) nested within therapists (1354 observations). To account for the resulting non-independence of assessments, and to prevent inflation of the effects, we added the patient as a random effect to the analyses, using the lme procedure for mixed model modeling from the nlme package of the R software (Pinheiro et al., 2012). Given that our focus in the present study is on the within-dyad effects, rather the between-dyad effects, we did not include therapist random effects (Falkenström et al., 2020). To measure the amount of variance in nonverbal synchrony due to random effects of the patient, we used the intraclass correlations (ICCs) index. The ICC of the patients' random effect was calculated as follows: $\mathrm{ICC} = \sigma^2_{\mathrm{patient}}/(\sigma^2_{\mathrm{patient}} + \sigma^2_{\mathrm{error}})$ where $\sigma^2_{\mathrm{patient}}$ and $\sigma^2_{\mathrm{error}}$ are the variances of the random effects of the

patient, and the error. We estimated random variance components for the patient, based on a model with only random intercept.

To examine the development of nonverbal synchrony over time, we evaluated the following trend models: linear, linear in log of time, and stability over time, either as fixed or random effects. We started with a model with only a fixed intercept and no random effects, and added a sequentially fixed effect of week and a random effect of week in therapy. Next, we examined the models with fixed and random linear effects of log of week. We used the likelihood ratio test and the Bayesian information criterion (BIC) to determine whether the inclusion of each term improved the model fit.

Aim 1: The effect of supportive techniques congruence and incongruence between patient and therapist on subsequent nonverbal synchrony at the sample level

To examine the effects of congruence and incongruence between patients and their therapists on the use of supportive techniques, on subsequent nonverbal synchrony, we conducted a multilevel model response surface analysis by polynomial regression (for more details, see Edwards, 2002; Edwards & Parry, 1993). This framework has been widely used for testing congruence and incongruence outside psychotherapy research (Shanock et al., 2010) and in psychotherapy research (Marmarosh & Kivlighan Jr, 2012). Following Shanock et al. (2010), the model included the following predictors: (a) patient supportive techniques rating, (b) therapist supportive techniques rating, (c) quadratic term formed by squaring the patient supportive techniques rating, (d) quadratic term formed by squaring the therapist supportive techniques rating, and (e) a cross-product term formed by multiplying the patient supportive techniques rating by the therapist supportive techniques rating. We controlled for all main effects. To establish a temporal relationship between the predictors and outcome, we used the predictors at time T-1 to predict outcome at time T, over the course of treatment. Due to the autoregressive nature of the data, we used the first order autoregressive (AR[1]) structure for the within-dyad residuals in all analyses. AR(1) was chosen in order to account for auto-regression in the dependent variable while avoiding a possible bias due to inclusion of the lagged dependent variable as a predictor in the model (Falkenström et al., 2017). Patients' and therapists' supportive techniques ratings were standardized. We focused on the within-dyad effects, the distance from the patients' and therapists' average over treatment, controlling for between-dyads effects. Based on Edwards (2002) and Edwards and Parry (1993), we used the following regression equation:

$$\begin{aligned}&\text{Nonverbal synchrony}_{it}\\&= b_0 + b_1\text{PSTw}_{i(t-1)} + b_2\text{TSTw}_{i(t-1)}\\&\quad + b_3\text{PSTw}^2_{i(t-1)} + b_4\text{PSTw}_{i(t-1)} \times \text{TSTw}_{i(t-1)}\\&\quad + b_5\text{TSTw}^2_{i(t-1)} + b_6\text{PSTb}_i + b_7\text{TSTb}_i\\&\quad + b_8\text{PSTb}_i^2 + b_9\text{PSTb}_i \times \text{TSTb}_i + b_{10}\text{TSTb}_i^2\\&\quad + b_{11}\log(t) + u_i + e_{it},\end{aligned}$$

where Nonverbal synchrony$_{it}$ is the nonverbal synchrony at session t of patient i; P/TSTb$_i$ represents the patient's/therapist's average supportive techniques over therapy (the between-dyad level); and P/TSTw $_{i(t\text{-}1)}$ represents the patient's/therapist's supportive techniques at the within-dyad level (difference from treatment average) at session $t-1$ of patient i; u_i and e_{it} are the random effects of patient i and the error at time t. All are independent and normally distributed, with mean 0 and variances $\sigma^2_{\text{patient}}$ and σ^2_{error} respectively.

We used the estimated coefficients of the regression model to calculate the estimates for two slopes and two curvatures along the response surface: (a1) = b_1+b_2, the slope of the line of congruence (patient's supportive techniques rating = therapist's supportive techniques rating); (a2) = b_3+b_4+b_5, the curvature along the line of congruence; (a3) = b_1-b_2, the slope of the line of incongruence (patient's supportive techniques rating = -therapist's supportive techniques rating); and (a4) = b_3-b_4+b_5, the curvature along the line of incongruence.

Aim 2 (Model 1–2): The effects of supportive techniques congruence and incongruence between patient and therapist on subsequent nonverbal synchrony, moderated by patients' levels of depression severity and PD levels

To examine whether depression severity may serve as a moderator in Model 1 of the effect of congruence and incongruence of supportive techniques, on subsequent nonverbal synchrony, we introduced the effect of depression severity into the model as an interaction between depression severity and each of the two slopes and two curvatures along the response surface. We carried out the same procedure with PD levels in Model 2. We used the

following equation:

$$\begin{aligned}&\text{Nonverbal synchrony}_{it}\\&= b_0 + b_1 PSTw_{i(t-1)} + b_2 TSTw_{i(t-1)} + b_3 PSTw^2_{i(t-1)}\\&\quad + b_4 PSTw_{i(t-1)} \times TSTw_{i(t-1)} + b_5 TSTw^2_{i(t-1)}\\&\quad + b_6 PSTb_i + b_7 TSTb_i + b_8 PSTb_i^2 + b_9 PSTb_i\\&\quad \times TSTb_i + b_{10} TSTb_i^2 + b_{11} Dp_i \times PSTw_{i(t-1)}\\&\quad + b_{12} Dp_i \times TSTw_{i(t-1)} + b_{13} Dp_i \times PSTw^2_{i(t-1)}\\&\quad + b_{14} Dp_i \times PSTw_{i(t-1)} \times TSTw_{i(t-1)} + b_{15} Dp_i\\&\quad \times TSTw^2_{i(t-1)} + b_{16} Dp_i \times PSTb_i + b_{17} Dp_i\\&\quad \times TSTb_i + b_{18} Dp_i \times PSTb_i^2 + b_{19} Dp_i \times PSTb_i\\&\quad \times TSTb_i + b_{20} Dp_i \times TSTb_i^2 + b_{21} Dp_i\\&\quad + b_{22} \log(t) + u_i + e_{it},\end{aligned}$$

where Dp_i is the depression severity of patient i. The interaction effect of each term for the within-subject supportive techniques agreement is calculated as: (a1) $b_{a1}=b_{11}+ b_{12}$ (a2) $b_{a2}= b_{13}+b_{14}+b_{15}$ (a3) $b_{a3}=b_{11}-b_{12}$ (a4) $b_{a4}= b_{13}-b_{14}+b_{15}$.

We estimated the region of significance of the depression severity-nonverbal synchrony association using the Johnson-Neyman technique (Hayes & Matthes, 2009).

Results

For nonverbal synchrony, the estimated variance of the patient's random effect was significant ($\sigma^2_{\text{patient}} = 0.83$, $p < .0001$, ICC = .39). A model of fixed effect of log of time was found to demonstrate the best model fit in predicting nonverbal synchrony, based on the BIC. Nonverbal synchrony significantly increased throughout the course of treatment ($\beta = 0.11$, $SE = 0.04$, $p = .005$) and was significantly higher than pseudosynchrony across all sessions [$t(1353) = 13.06$, $p < .001$, Cohen's $d = 0.36$]. Repeating our analyses without time in the models resulted in the same pattern of results (see Tables S5–S7 in the Supplements).

Aim 1: The effect of supportive techniques congruence and incongruence between patient and therapist on subsequent nonverbal synchrony at the sample level

We used response surface analysis to examine the effects of patient and therapist congruence and incongruence of supportive techniques levels on subsequent nonverbal synchrony. In the course of this work, we examined the within-dyad agreement effects, while controlling for between-dyads effects.

Findings were significant for the line of congruence but not for the line of incongruence (Table I). We found a significant slope along the line of congruence ($a_1 = 0.08$, $p = .03$), suggesting that congruence between patient and therapist on higher levels of within-patient supportive techniques was associated with higher nonverbal synchrony at the subsequent session. In Figure 1, the x-axis represents patient-rated supportive techniques, the y-axis therapist-rated supportive techniques, and the z-axis the expected value of subsequent nonverbal synchrony according to the model. The line of congruence of patient and therapist supportive techniques ratings (where patient's supportive techniques rating = therapist's supportive techniques rating) extends from the closest to the farthest corners of the plane. The slope of the response surface along the line of congruence shows the effect of congruence at high and low levels of patient and therapist supportive techniques. The slope of the line of congruence shows that, on average, nonverbal synchrony is higher when patient and therapist ratings of supportive techniques are high than when they are low. In other words, congruence on higher levels of supportive techniques predicted higher nonverbal synchrony at the subsequent time point than did congruence on a lower level of supportive techniques.

Regarding the line of incongruence, where within-patient and therapist supportive technique ratings were opposite (patient supportive techniques rating = -therapist supportive techniques rating), both the slope and curvature were non-significant ($a_3 = 0.07$, $p = 0.08$), ($a_4 = 0.07$, $p = 0.09$). These findings lend no support to the ability of incongruence on supportive techniques or its direction to predict nonverbal synchrony.

Aim 2 (Model 1–2): The effects of supportive techniques congruence and incongruence between patient and therapist on subsequent nonverbal synchrony, moderated by patients' levels of depression severity and PD levels

The association between the two moderators, depression severity and PD levels, was found to

Table I. Response surfaces for patient- and therapist-rated supportive techniques and subsequent nonverbal synchrony.

	Between		Within	
Effect	Coefficient	*P*	Coefficient	*p*
a_1: Slope along $x = y$	−0.26	.14	0.08	.03
a_2: Curvature along $x = y$	−0.14	.50	0.03	.22
a_3: Slope along $x = -y$	0.29	.15	0.07	.08
a_4: Curvature along $x = -y$	−0.15	.50	0.07	.09

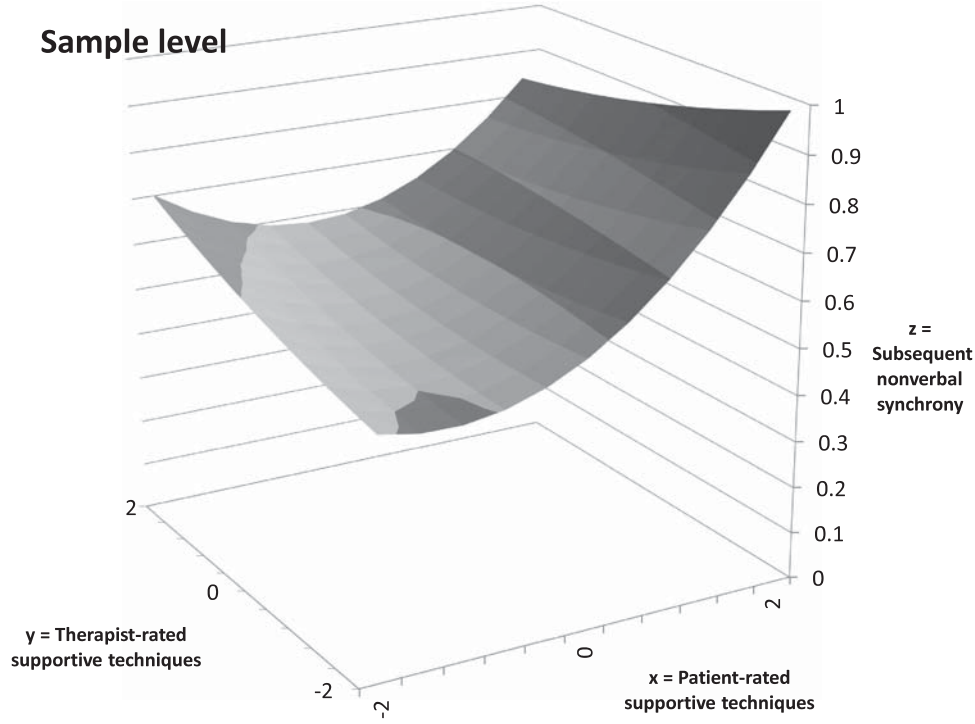


Figure 1. Congruence between patient and therapist ratings of supportive techniques and subsequent nonverbal synchrony.

be moderately correlated ($r = 0.37$, $P < .001$), which is consistent with previous literature showing similar results (Lozano & Johnson, 2001; Santor et al., 1997). Correlations between variables at the between- and within-individual levels appear in Tables S1 and S2, in the Supplements. Whereas Model 1 focused on depression severity, Model 2 focused on PD levels. We examined depression severity and PD levels as moderators of the effect of congruence on supportive techniques on subsequent levels of nonverbal synchrony. To assess the differential effects of depression severity and PD levels on the ability of patient-therapist congruence of supportive techniques to predict subsequent nonverbal synchrony, we added interaction terms with depression severity in Model 1, and with PD levels in Model 2, to all the terms of the original model, and examined the linear combinations of the interaction effects. We found a significant interaction between depression severity and the slope along the line of incongruence ($b_{a3} = -0.03$, $p = .003$) (Table II). Similarly, we found a significant interaction between PD levels and the slope along the line of incongruence ($b_{a3} = -0.004$, $p = .03$) (Table III).

To shed light on these significant interactions, we derived a separate response surface for low levels (that is, one standard deviation (SD) below their mean value) and high levels (one SD above their mean value) of depression severity (Table S3 in the Supplements), and for low and high levels of PD (Table S4 in the Supplements). Figure 2a shows that, for patients with low levels of depression

Table II. Moderating effect of depression severity on the association between patient- and therapist-rated supportive techniques and subsequent nonverbal synchrony.

	Between		Within	
Moderated Effect	Coefficient	*P*	Coefficient	*p*
b_{a1}: Slope along $x = y$	0.004	.94	−0.01	.29
b_{a2}: Curvature along $x = y$	0.01	.86	−0.01	.08
b_{a3}: Slope along $x = -y$	0.06	.27	−0.03	.003
b_{a4}: Curvature along $x = -y$	0.03	.75	−0.008	.46

Table III. Moderating effect of personality disorders on the association between patient- and therapist-rated supportive techniques and nonverbal synchrony.

	Between		Within	
Effect	Coefficient	*P*	Coefficient	*p*
b_{a1}: Slope along $x = y$	−0.014	.13	−0.001	.50
b_{a2}: Curvature along $x = y$	0.004	.73	0.001	.57
b_{a3}: Slope along $x = -y$	0.009	.32	−0.004	.03
b_{a4}: Curvature along $x = -y$	−0.02	.07	0.000	.82

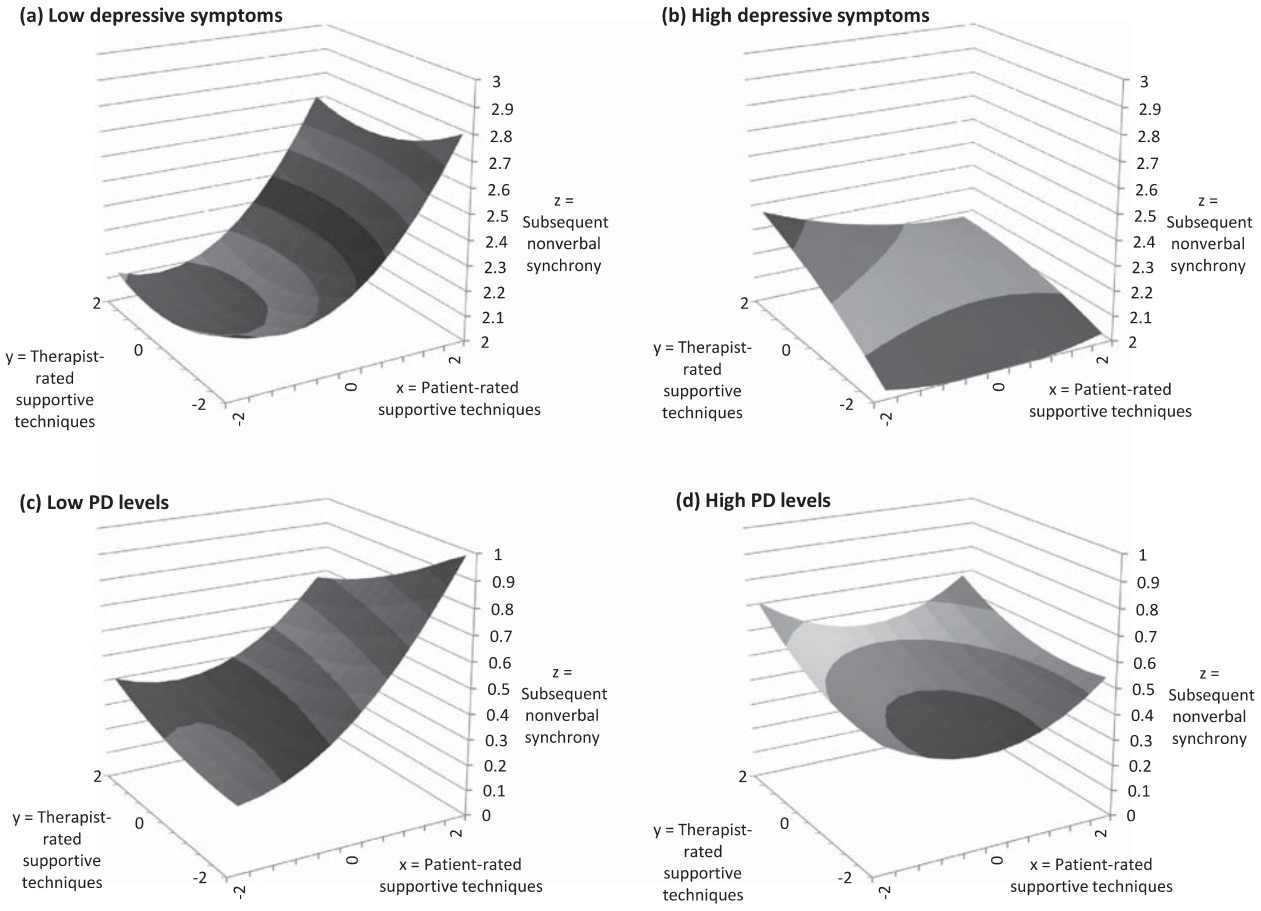


Figure 2. Incongruence between patient and therapist ratings of supportive techniques and subsequent nonverbal synchrony. (a) low levels of depressive symptoms; (b) high levels of depressive symptoms; (c) low PD levels; (d) high PD levels.

severity, when they reported higher levels of supportive techniques than did their therapists, the nonverbal synchrony was higher at the next session, than vice versa (i.e., when the therapists reported higher levels of supportive techniques than did their patients). The effect of supportive techniques on subsequent nonverbal synchrony was not significant for patients with high levels of depression severity (Figure 2b). Similarly, Figure 2c shows that for patients with low PD levels, when they reported higher levels of supportive techniques than did their therapists, the nonverbal synchrony was higher at the next session, than vice versa (i.e., when the therapists reported higher levels of supportive techniques than did their patients). The effect of supportive techniques on subsequent nonverbal synchrony was not significant when patients displayed high PD levels (Figure 2d). According to the region of significance calculation (Hayes & Matthes, 2009), from an HRSD score of 19 (HRSD-17; Hamilton, 1967), which indicates a severe level of depressive symptoms (National Collaborating Centre for Mental Health, 2010), the effect of supportive techniques on subsequent nonverbal synchrony was not significant. Standardized coefficients of all the effects appear in the supplements (Tables S8–S10).

Discussion

The present study sought to examine the therapeutic processes that are set in motion following the use of supportive techniques in therapy. Specifically, we examined whether greater use of supportive techniques can predict higher levels of nonverbal synchrony between patient and therapist at a subsequent session. We hypothesized that patient and therapist congruence on higher levels of supportive techniques is associated with subsequent higher levels of nonverbal synchrony. The findings support this hypothesis and show that congruence on greater use of supportive techniques predicts higher levels of nonverbal synchrony at the next session. We further hypothesized that the effect of supportive techniques on subsequent levels of nonverbal synchrony is moderated by patient factors, specifically

depression severity and PD levels. The findings show that the effect of supportive techniques on subsequent levels of nonverbal synchrony is significant in patients with low levels of depression severity and with low levels of PD, but not in patients with high levels.

The first aim of the present study focused on the sample level and examined the association between the patient-therapist congruence on the use of supportive techniques and subsequent nonverbal synchrony. The findings were consistent with our hypothesis, suggesting that when the patient and therapist congruently rated greater use of supportive techniques, they showed higher levels of nonverbal synchrony at the next session. This finding is consistent with previous literature suggesting that supportive techniques contribute to the closeness between the patient and therapist and help bring them to a shared experience (Barber et al., 2001; De Jonghe et al., 1994; Wachtel, 2011). Similarly, the findings are consistent with previous reports suggesting that higher levels of nonverbal synchrony are associated with a better therapeutic relationship between the patient and therapist (Altmann et al., 2020; Ramseyer & Tschacher, 2011) and that patient-therapist congruence is important for positive interpersonal processes (Chui et al., 2020; Marmarosh & Kivlighan Jr, 2012; Rubel et al., 2018). Assessing the congruence between patient and therapist and evaluating subsequent synchrony may have helped explain the different findings from those of Prinz et al. (2021), who reported a negative association between resource activation and nonverbal synchrony at the same session. Supportive techniques may need some time to manifest in observable behavior, such as nonverbal synchrony, and the patient-therapist congruence may act as a further important factor in this association.

The second aim of the present study focused on individual differences between patients in the association between the patient-therapist congruence of the use of supportive techniques and subsequent nonverbal synchrony. We focused on patient factors that might impair the ability of the patient and therapist to be in a shared experience and synchronize, specifically depression severity and PD levels. Both patient factors, depression severity and PD levels, were found to be significant moderators and in the same direction. The findings were consistent with our hypothesis regarding the direction of the moderation, suggesting that for patients with low levels of depression severity or low levels of PD at the beginning of treatment, supportive techniques result in subsequent higher levels of nonverbal synchrony. Yet, for patients with high levels of depression severity or high levels of PD at the beginning of treatment, the effect of supportive techniques on nonverbal synchrony was not significant. This finding is consistent with previous research showing that high levels of depression severity and PD are related to difficulties in forming and maintaining the therapeutic relationship between the patient and therapist (Clarkin & Levy, 2004; Flückiger et al., 2020; Muran et al., 2009). This finding is consistent also with previous literature showing that therapists working with patients with higher baseline symptom severity experience more difficulties using specific interventions and techniques, measured by therapist reports (Wolf & Goldfried, 2014) and external coders (Boswell et al., 2013; Imel et al., 2011).

We found individual differences in the way incongruence between patient and therapist on the use of supportive techniques was associated with subsequent nonverbal synchrony. Although we had clear hypotheses for the direction of the effect, the nature of the effect (congruence and incongruence) was exploratory, because of the lack of previous research on the subject. The results indicate that in patients with low levels of depressive symptoms or low levels of PD, nonverbal synchrony at the next session was higher when patients rated the levels of supportive techniques higher than did their therapists, rather than vice versa. A potential *post hoc* interpretation of the findings is that, for patients with low levels of depressive symptoms or PD, the supportive techniques were perceived in a magnified way because of the great impact these techniques had on these patients, their ability to be synchronized with their therapists, and on their therapeutic process. The therapists of these patients may have underestimated how meaningful the supportive techniques were for them, rating the use of techniques lower than their patients did. This is consistent with previous findings showing that patients with mild symptoms of depression benefited from supportive therapy (Cuijpers et al., 2012).

If future studies support the current findings and further suggest a causal relationship between supportive techniques and nonverbal synchrony, this study may have important clinical and empirical implications. The findings illustrate that greater use of supportive techniques in therapy may result in higher levels of nonverbal synchrony between patients and their therapists. This emphasizes the importance of supportive techniques in bringing both patient and therapist to be synchronized and in a shared experience, contributing to their ability to work collaboratively together (De Jonghe et al., 1994; Misch, 2000). Current literature demonstrates the value of nonverbal synchrony between patient and therapist in psychotherapy and its relation to better therapeutic processes and outcome (Altmann et al., 2020; Ramseyer & Tschacher, 2011; Wiltshire

et al., 2020). The present study suggests that it may be in the therapist's power to influence and change the nonverbal synchrony within the therapy setting. In the case of a patient-therapist dyad where nonverbal synchrony is lacking, the therapist can choose to make greater use of supportive techniques to increase the nonverbal synchrony between them and bring them to a shared experience. The findings further show that this is true for patients with lower levels of symptoms, but not for patients with higher levels. Therapists, indeed, experience more difficulties in the therapeutic process with patients with higher levels of symptoms (Wolf & Goldfried, 2014), and these patients may require that the therapist implement additional interventions, such as expressive techniques (Høglend et al., 2007).

Furthermore, if future studies show a causal association between supportive techniques and nonverbal synchrony, the current findings can assist in implementing an automatic in-session feedback system that would provide feedback to therapists regarding their use of techniques and its relation to the ability of the patient and therapist to be in a shared experience (Imel et al., 2019; Lutz et al., 2019). This will provide information on within-dyad development of the relationship between the patient and therapist throughout the session, in relation to the therapist's use of techniques. This kind of feedback enables a monitoring system to indicate when the interaction between the patient and therapist is becoming more synchronized and when it is becoming less so, so that therapists can receive real-time feedback on whether the techniques they have used are effective in bringing the patient and therapist to be more synchronized (Kleinbub et al., 2020). Innovative developments in real-time computer-based feedback systems in psychotherapy have been shown to improve therapists' efficacy (Imel et al., 2019). For example, if therapists had real-time feedback to assist them in detecting that they and the patient have low levels of synchrony, this information could be crucial in directing them to use more supportive techniques and advance the therapy more efficiently (Lutz et al., 2019).

The present study has several limitations. Although the sample size is moderate compared to other samples used in studies examining nonverbal synchrony, it might still not be large enough to detect small effects. We used designated measures to assess supportive techniques, depression severity, and PD levels. For supportive techniques, we used patient and therapist ratings, and for depression severity and PD levels we used clinical interviews. There is much debate in the literature about how to measure these factors, especially PD (Huprich & Bornstein, 2007; Wright et al., 2012). Future studies should examine whether our findings can be replicated using other measures. Although using both patient and therapist ratings of supportive techniques is expected to be more reliable than only patient or therapist ratings, it may still provide a different type of information than can be yielded using observer-ratings, which should be examined in future studies. The present study used polynomial regression and response surface analysis to examine agreement and disagreement, but there are other ways to examine these constructs, and these should be the focus of future studies. The association between the two moderators, depression severity and PD levels at baseline, is of medium size, which is consistent with previous literature showing similar results (Lozano & Johnson, 2001; Santor et al., 1997). The reliability for the therapist version of the MULTI is rather low, which is consistent with previous literature showing similar results (Solomonov et al., 2019). Because this trial has yet to be terminated, we could not use treatment condition, adherence and outcome. It would be interesting for future studies to examine a mediation model including the effects of supportive techniques and nonverbal synchrony on treatment outcome. Last, the present findings are specific to patients with MDD. Future studies should examine whether the findings can be generalized to additional clinical populations.

This is the first study to show the effect of supportive techniques used by therapists on nonverbal synchrony at the subsequent session. The findings suggest that, at the sample level, patient and therapist congruence of greater use of supportive techniques predicts higher levels of nonverbal synchrony. The findings further suggest that the effect of supportive techniques on nonverbal synchrony is moderated by patient factors, specifically depression severity and PD levels.

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Supplemental data

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